

Polynomials – 5

Integer coefficients, divisibility and irreducibility

Homework

8 сентября 2026 года

Theoretical minimum. Gauss's lemma and content; the criteria of Eisenstein, of reduction modulo a prime and of Perron; the divisibility $a - b \mid f(a) - f(b)$; integer-valued polynomials and the binomial basis; conjugate roots of an irreducible rational polynomial.

1. (IMC) Let p be a polynomial with integer coefficients and let $a_1 < a_2 < \dots < a_k$ be integers.
 - a) Prove that there exists $a \in \mathbb{Z}$ such that $p(a_i)$ divides $p(a)$ for every i .
 - b) Does there exist $a \in \mathbb{Z}$ such that the product $p(a_1)p(a_2)\dots p(a_k)$ divides $p(a)$?
2. (IMC) Let f be a rational function, that is, a quotient of two real polynomials, and suppose that $f(n)$ is an integer for infinitely many integers n . Prove that f is a polynomial.
3. (Proposed problems) Find all polynomials $P \in \mathbb{Z}[X]$ such that $a^2 + b^2 + c^2 \mid P(a) + P(b) + P(c)$ for all integers a, b, c .
4. (Proposed problems) Determine all non-constant polynomials $P \in \mathbb{Z}[X]$ such that $P(p)$ is square-free for every prime p .
5. (Proposed problems) Let $f \in \mathbb{Z}[X]$ with $\deg f \geq 1$ and let k be a positive integer. Show that there are infinitely many positive integers n such that $f(n) = d_1 d_2 \dots d_k d_{k+1}$ with $1 \leq d_1 < d_2 < \dots < d_k < n$.
6. (VJIMC 2013) Prove that there is no polynomial P with integer coefficients such that $P(\sqrt[3]{5} + \sqrt[3]{25}) = 5 + \sqrt[3]{5}$.

7. (Сибирская 2024) Does there exist a polynomial $p(x)$ with integer coefficients such that $p(3 + \sqrt{5}) = 3 + \sqrt{5}$ and
- a) $p(3 - \sqrt{7}) = 4 - \sqrt{7}$?
 - b) $p(3 - \sqrt{7}) = 5 - \sqrt{7}$?
8. (Iranian competition) Find all polynomials f with rational coefficients such that $f(x)$ is irrational whenever x is irrational.
9. (Putnam 2007) A repunit is a positive integer all of whose decimal digits are ones. Find all polynomials f with real coefficients such that $f(n)$ is a repunit whenever n is.
10. (Putnam 2016) Find the smallest positive integer j such that for every polynomial $p(x)$ with integer coefficients and every integer k the number $p^{(j)}(k)$, the j -th derivative of p at k , is divisible by 2016.